# Contamination Model: Team Update

Oct 24, 2026 · @Ishan SIngh

The prototype runs end to end and has now been tested on data we did not write. On a published industrial-scale simulation (IndPenSim) it caught all 10 equipment faults, a median of 1.5 h after they started, with 1 false alarm in 30 normal batches and nothing mistaken for contamination. When a simulated contaminant was added to those realistic batches, it caught 27 of 30 contaminants that changed respiration by about 20%, within about 19 h. It has not yet been tested on real MaverickX data or on a real contamination.

## In plain words

The system is a second pair of eyes on every enzyme batch. Today MaverickX spots contamination by smell, colour and experience, often after the batch is already lost. Our system watches the routine measurements (pH, temperature, dissolved oxygen, biomass and enzyme activity) and answers four questions for the scientist:

1. **Is this batch behaving unusually?** It compares the batch with what normal batches looked like at the same hour.
2. **What is the most likely reason?** Contamination, slow but normal growth, a change in raw materials or settings, a faulty sensor, or a failing pump, valve or cooling supply.
3. **What should we check next?** It suggests the quickest, cheapest test that tells the likely reasons apart.
4. **Where might it have come from?** It looks for a medium lot, inoculum lot or bioreactor that the contaminated batches have in common.

It supports the scientist's decision; it never discards a batch or declares contamination on its own. Only a real lab test can mark contamination as confirmed.

## How it works

Each batch goes through six steps, from raw readings to a one-page investigation report.

1. **Check the data.** Missing, impossible, duplicated or wrong-unit readings are flagged, never deleted or filled with zero, so no evidence is lost silently.
2. **Learn what normal looks like.** From batches the scientists approve as normal, it learns the typical value of each measurement at each hour and how much it usually varies. This reference is then frozen and versioned, so slow drift can't quietly become the new normal; a refresh workflow proposes newly approved batches for a scientist to sign off.

3. **Spot unusual behaviour.** Every new reading is compared with the reference. Five simple, explainable detectors raise alerts: one reading far from normal, a sustained drift, an accumulating shift, two or more linked signals ($CO_2$ output, oxygen uptake, dissolved oxygen) changing together, and a controlled variable drifting while its pump or cooling does not respond. How tight the limits are is set by a false-alarm budget: at most 5% of known-normal batches would have alerted. Each alert records what changed, by how much, when, and which other signals moved with it.
4. **Suggest explanations.** A rule book that scientists can read and edit links patterns of evidence to explanations. For example, extra biomass plus extra oxygen use plus a more acidic culture points to contamination; one signal drifting alone points to a faulty probe. Explanations get a transparent score, not a probability. When the evidence is thin, the honest answer is "unresolved, too early to tell".
5. **Recommend the next check.** It picks the check that best separates the leading explanations (e.g. microscopy versus a probe check), and keeps a contamination check on the list until contamination is ruled out.
6. **Trace the source.** It compares how often contamination happened in batches that used each medium lot, inoculum lot and bioreactor versus batches that did not. A resource that stands out is reported as a lead to investigate, not proof.

When a scientist records a test result (say, microscopy finds foreign cells), the assessment updates straight away.

## Results so far

We tested on three kinds of data. On our own simulator (20 fresh campaigns, 500 test batches not used for tuning) it caught every contaminated batch and named the right cause for 295 of the 300 problem batches by the end of the batch. Because we wrote both the simulator and the rules, those numbers show the software works as designed; the external tests below are the more honest measure.

### On our own simulator

| What we measured | Result |
|---|---|
| Contaminated batches caught | 120 of 120 |
| Warning before the visual/odour check | about 7.5 h earlier (median; 5–9 h across campaigns) |
| Sensor faults caught (spike or drifting probe) | 120 of 120 |

| What we measured | Result |
|---|---|
| Normal-category batches flagged | 81 of 260 (60 of them are genuinely slow-growing batches) |
| Right explanation, 8 h after the first alert | 241 of 300 (6 wrong, the rest "too early to tell") |
| Right explanation, end of batch | 295 of 300 (3 unresolved, 2 wrong) |
| Shared medium lot found as the top lead | 20 of 20 campaigns |
| Campaigns with a false strong lead | 1 of 20 (a real chance cluster) |

At the very first alert the system mostly says "too early to tell", on purpose: one signal moving alone fits contamination, a faulty probe and normal variation alike. Full numbers and method are in `docs/mvp-results.md` in the repo.

## Tested on data we did not write

| Test | Result |
|---|---|
| IndPenSim: 10 batches with equipment faults (aeration, feed, base pump, cooling) | All 10 caught, median 1.5 h after the fault started; all 4 pump and cooling faults correctly called an equipment fault |
| IndPenSim: 30 normal held-out batches | 1 flagged (one temperature reading); none called contamination |
| IndPenSim + simulated contaminant, +20% change in respiration | 27 of 30 caught, about 19 h after it started (was 5 of 30 before the linked-signal detector) |
| Same, +40% | 30 of 30 caught at about 13 h; contamination named as the cause in 29 of 30 |
| Same, +10% | 1 of 30: below what the model can see on realistic variation |
| Real erythromycin plant, 163 later batches (no labels) | 26 flagged (16%); flagged batches had lower final potency ($p = 0.04$); none called contamination |
| Standard multivariate method (PCA) on IndPenSim, same false-alarm budget | 7 of 10 faults: our simpler detectors did better and explain themselves |

The first sign of a simulated contaminant was almost always $CO_2$ output, so off-gas data is the most useful thing to ask MaverickX for. Every count in the repo now comes with a 95% range, e.g. "10/10 (72–100%)": small samples should not be over-read. Details: `docs/external-validation.md`.

## The dashboard

Scientists use a web dashboard; no coding needed. A sidebar picks the dataset (the synthetic demo, IndPenSim, the erythromycin plant, or data loaded through the dashboard) and the false-alarm budget (1–20%).

| Page | What a scientist can do |
|---|---|
| Campaign overview | See every batch at a glance: which are flagged, the likely cause, whether contamination is confirmed, and which other batches are linked |
| Batch investigation | See each measurement against the normal band with alerts marked; replay the batch hour by hour; see how the assessment developed (when the first alert came, when each explanation first led); read the explanations, evidence and next checks; record a test result; mark each alert as a real issue or a false alarm; download the report |
| Detection timing | For batches with a known start of the problem: hours to the first alert and how much earlier than today's check |
| Source tracing | See which lots or bioreactors the contaminated batches share, drawn as a graph, with the statistics behind each lead |
| Value estimate | A yearly savings estimate from the plant's own figures (placeholders until MaverickX gives theirs) |
| Load your data | Upload plant exports and a mapping file, see a readiness report, and monitor them |
| How it works | A plain-language explanation of the method and its limits |

Check results and alert feedback are saved per dataset and reload next time (download them before a restart on Streamlit Cloud).

## Limits

The results show the software works as designed and holds up on realistic external data. They do not yet show it works on MaverickX's real process or on a real contamination.

- **No real contamination yet.** Our own α-amylase simulator (with E. coli as the example contaminant), the IndPenSim simulation and the erythromycin plant contain no real contamination. The contaminant's effects in both contamination tests are our assumptions.
- **The rule book is a draft.** Every rule, weight and suggested check was written by us and needs review by MaverickX scientists. Check times and costs are placeholders.
- **It flags unusual batches, not organisms.** It never identifies a species from pH or oxygen readings, and its scores are not probabilities.
- **Small contaminants are invisible, and false alarms remain.** On realistic variation a contaminant that changes respiration by about 10% is not detected; about 20% is caught in most batches. At +20% the model usually alerts but leaves the cause "unresolved" until the 12-hourly biomass result confirms it. On the real plant, 16% of later batches were flagged, many for respiration changes that did not hurt the product.
- **Source tracing sees only what is recorded:** medium lot, inoculum lot and bioreactor. Water, transfer lines and environmental swabs aren't included, and with small campaigns a chance cluster can look like a real source.
- **It investigates; it doesn't predict.** Predicting contamination before it starts would need historical data with confirmed cases.

## What we need from the team

The prototype is ready to show; the deliverables are due `Oct 28, 2026`.

- [ ] Review the rule book (`src/contamination/knowledge/catalogue.yaml`): are the contamination signs and weights biologically sensible? (Aisha, Beren)
- [ ] Confirm the proxy choice: α-amylase with E. coli, or lipase with sulfate-reducing bacteria
- [ ] Send MaverickX the one-page data request (docs/MaverickX-Data-Request.pdf), stressing off-gas $CO_2$/oxygen data, controller outputs (pump and cooling) and confirmed contamination cases
- [ ] Ask MaverickX which tests they can actually run, with real turnaround times and costs, to replace our placeholders
- [ ] Ask MaverickX how many normal batches in 100 they can afford to review (sets the false-alarm budget)
- [ ] Use the dashboard and the example report in the one-pager and technical document

- [ ] Share with Pranav and Shea and the validators for feedback

## How to run it

Everything is in the triple-ignition-challenge repo and needs Python 3.11 or newer.

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.lock && pip install -e ".[dev]"
python -m contamination.demo      # generates the simulated batches and
reports
streamlit run app/dashboard.py    # opens the dashboard at
http://localhost:8501
```

The README has the code map, and `docs/example-investigation-report.md` shows a full report for one contaminated batch. The IndPenSim and erythromycin datasets ship with the repo and open from the dashboard's dataset picker; `docs/technical-document.pdf` explains the whole model in detail.